

HASSAN KARUNGWA

Full-Stack Software Engineer | Backend-Focused | SaaS & Systems Architecture

Homabay, Kenya · Open to Remote

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PROFESSIONAL SUMMARY

Full-Stack Software Engineer specializing in building production-oriented, multi-tenant SaaS systems with a strong backend focus in Java and Spring Boot and hands-on expertise across React, Next.js, TypeScript, PostgreSQL, and cloud-ready infrastructure.

Experienced in designing domain-driven systems using Clean Architecture, DDD, SOLID principles, modular monoliths, aggregates, domain events, tenant isolation, secure authentication, REST APIs, and typed API contracts. Strong experience implementing business-critical workflows involving payments, billing, financial ledgers, leases, and transactional integrity.

Built and deployed RentManager, a multi-tenant property-management SaaS platform spanning web and mobile applications, with backend, frontend, and mobile systems designed to work as an integrated product.

Focused on building systems that are maintainable, secure, testable, scalable, and suitable for real-world production environments.

TECHNICAL SKILLS

Languages: Java · TypeScript · JavaScript · SQL

Backend Engineering: Spring Boot · Spring Data JPA · Hibernate · REST APIs · OpenAPI · JWT Authentication · Application Services · Domain Events · Aggregates

Frontend Engineering: React · Next.js · TypeScript · React Query · Responsive Web Applications · Typed API Integration

Mobile Development: Android · iOS · Mobile Application Development

Architecture & Engineering: Clean Architecture · Domain-Driven Design (DDD) · SOLID · System Design · Modular Monoliths · Multi-Tenant SaaS · Event-Driven Architecture · Domain Modeling

Databases & Data: PostgreSQL · SQL · Flyway · Database Migrations · Tenant-Scoped Persistence

Infrastructure & Distributed Systems: Docker · Redis · Kafka · Containerized Services

Payments & Financial Systems: M-Pesa Daraja · Payment Gateway Integration · Billing Workflows · Financial Ledgers · Idempotent Transactions · Transaction Safety

Security & Reliability: Authentication · Authorization · Tenant Isolation · Secure API Design · Validation · Error Handling · Transaction Integrity

Testing & Development Practices: Unit Testing · Integration Testing · Database-Backed Testing · API Contract Testing · Git · GitHub · API Documentation

PROFESSIONAL EXPERIENCE

Full-Stack Software Engineer — Independent / Freelance

February 2023 – Present

- Design and develop production-oriented software systems across backend, web, and mobile applications, with a strong focus on scalable architecture and maintainable domain models.
- Architect and implement multi-tenant SaaS systems with tenant-scoped persistence, secure authentication, UUID-based identities, and clearly defined domain boundaries.
- Apply Domain-Driven Design, Clean Architecture, SOLID principles, aggregates, domain events, and application services to keep business logic independent from infrastructure concerns.
- Build REST APIs and maintain OpenAPI contracts to establish reliable integration between backend and frontend applications.
- Develop financial and billing workflows with emphasis on transaction safety, idempotency, and consistency.
- Work with PostgreSQL, Flyway, Docker, Redis, and Kafka-style event infrastructure to build production-oriented application environments.
- Develop automated unit and integration tests, including database-backed tests, to validate domain behavior and API contracts.
- Build full-stack applications using Java/Spring Boot on the backend and React/Next.js/TypeScript on the frontend.

- Develop cross-platform mobile applications for Android and iOS, extending SaaS functionality beyond the web platform.

SELECTED PROJECTS

RentManager — Multi-Tenant Property Management SaaS

Full-Stack Developer / Software Architect

Technology: Java 21 · Spring Boot · PostgreSQL · Flyway · Docker · Redis · Kafka · React · Next.js · TypeScript · React Query · Clerk

- Architected and deployed a multi-tenant property-management SaaS platform covering properties, units, tenants, leases, payments, maintenance, subscriptions, and organization management.
- Designed tenant-aware domain architecture using UUID identifiers, tenant-scoped persistence, domain boundaries, aggregates, and domain events.
- Implemented lease lifecycle workflows including approval, activation, termination, and renewal, with explicit workflow validation.
- Built M-Pesa-based payment and billing capabilities with idempotent transaction handling and separation between business rules and infrastructure.
- Developed the Next.js web application and integrated it with the Spring Boot backend through typed API contracts, Clerk-based authentication, and React Query data flows.
- Developed a companion mobile application for both Android and iOS, extending the platform's capabilities to mobile users.
- Implemented PostgreSQL database migrations with Flyway and containerized application infrastructure using Docker.
- Applied centralized error handling, API documentation, automated testing, and production-oriented engineering practices.
- Structured the system to support continued evolution toward a scalable SaaS product.

Repositories: [rentmanager-backend](#) · [rentmanager-frontend](#) · [rentmanager-mobile](#)

Kenya Keys Sponsorship System

- Designed a UUID-based architecture with secure, tenant-isolated data modeling and domain-oriented backend structure.
- Implemented a financial ledger with idempotent transaction handling to improve financial consistency and prevent duplicate operations.
- Applied Clean Architecture and Domain-Driven Design principles to separate core business logic from infrastructure concerns.

EDUCATION

Tom Mboya University — Homabay, Kenya

Bachelor of Science (BSc), Computer Science · Expected Graduation: 2028

PROFESSIONAL FOCUS

Backend & Full-Stack Software Engineering · SaaS Platform Engineering · Java & Spring Boot Development · Multi-Tenant Systems · Domain-Driven Design · Software Architecture & System Design · Payment & Financial Systems · API & Integration Engineering · React / Next.js Development · Mobile Application Development · Production-Oriented Engineering

LINKS

GitHub: github.com/hassan200503 · Portfolio: hassan200503.github.io